

Comparative Assessment: CRISLTLSUM vs. CNN/DAILYMAIL

Deployment context: Ecuador — Humanitarian Crisis Social Media Analysts

Deployment Context

Use case: Timeline generation for coordination and evaluation of response efforts across time published in social media. The data would be tweets from natural disaster domain generated in Ecuadorian Spanish.

Target population: Latin American data and social analysts working at NGO and international organizations (e.g. Red Cross, UN OCHA) who seek to find how affected populations' needs evolve across time. They would be fluent in American English and Spanish, potentially speakers of different accents of Latin American Spanish.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
CRISLTLSUM	HIGH	1.3 / 5
CNN/DAILYMAIL	HIGH	1.0 / 5

Dimension Scores Comparison

Dimension	CRISLTLSUM	Rating	CNN/DAILYMAIL	Rating	Δ
Input Ontology (IO)	1/5	Serious concern	1/5	Serious concern	0
Input Content (IC)	1/5	Serious concern	1/5	Serious concern	0
Input Form (IF)	1/5	Serious concern	1/5	Serious concern	0
Output Ontology (OO)	2/5	Significant gaps	1/5	Serious concern	+1
Output Content (OC)	1/5	Serious concern	1/5	Serious concern	0
Output Form (OF)	2/5	Significant gaps	1/5	Serious concern	+1
Average	1.3		1.0		+0.3

Δ = CRISLTLSUM score – CNN/DAILYMAIL score. Positive = regional benchmark scores higher.

Overall Summaries

CRISLTLSUM (risk: HIGH)

CrisisLTLSum is fundamentally misaligned with the Ecuador humanitarian crisis social-media analyst deployment along five of six validity dimensions. Its taxonomy excludes all of Ecuador's priority crisis types (volcanic, seismic, landslide, flooding, displacement, complex emergencies); its content is entirely US/Australian English Twitter; its preprocessing pipeline is English-locked and the authors themselves flag non-portability [Q119]; its annotators were restricted to English-speaking countries [Q61], not the inter-operational humanitarian teams the deployment requires; and its evaluation rubric is anchored to English coherence [Q153]. The only dimension scoring above 2 is Output Form (surface text-based modality matches), and even there ROUGE and the human rubric require Spanish/genre adaptation. The benchmark's methodology — semi-automated cluster-then-refine, MTurk QC stack, four-axis summary rubric — is auditable and re-implementable as a template, but the artifact itself transfers essentially nothing usable to the deployment context.

CNN/DAILYMAIL (risk: HIGH)

The CNN/Daily Mail Reading Comprehension benchmark is categorically misaligned with the Ecuador Humanitarian Crisis Tweet Analysis deployment across all six validity dimensions, all rated HIGH priority in the elicitation. The benchmark defines a single Cloze-style entity-fill-in task on formal English news articles averaging 763 tokens, with algorithmically-generated labels reflecting anglophone editorial conventions of two Western news outlets. The deployment requires temporally ordered narrative timeline generation from informal Ecuadorian Spanish tweets (with Kichwa entity tokens) anchored to parroquia/cantón/provincia administrative levels, validated by inter-operational humanitarian teams. Empirical sampling of the HuggingFace dataset confirms zero Spanish content, zero Latin American geographic coverage, zero humanitarian crisis content, and no human annotation. Every dimension scored 1. No partial reuse pathway is supported by the evidence — the benchmark and deployment share essentially no structural, content, or task overlap.

Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

CRISISLTLSUM — 1/5 (Serious concern) [high confidence]	CNN/DAILYMAIL — 1/5 (Serious concern) [high confidence]
The benchmark restricts itself to four crisis domains — wildfire, local fire, traffic, and storm [Q122] — and explicitly defines events as 'local' incidents 'bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period' [Q12]. The Ecuador deployment's priority crisis types are volcanic eruptions, earthquakes, landslides, coastal/riverine flooding, mass displacement, and complex humanitarian emergencies — none of which appear in the benchmark taxonomy. The elicitation explicitly identifies large-scale humanitarian crises that 'exceed local response capacity' as the operational priority, structurally incompatible with the benchmark's local/short-duration framing. Storms have only partial overlap, and even there the benchmark's storm category was selected for North American contexts (Texas/Florida hurricanes, NSW storms) [Q124], not Andean lahars or tropical Pacific precipitation. An independent Ecuadorian Twitter monitoring system identified four distinct categories — volcanic, telluric (seismic), fires, and climatological — confirming that Ecuador's operational taxonomy diverges fundamentally [WEB-18].	The benchmark defines a single unified task — Cloze-style entity fill-in over a context document estimating $p(a c,q)$ [Q10] — and explicitly scopes itself to single-document linguistic relationship detection while excluding world knowledge and co-occurrence statistics [Q11, Q17]. There is no humanitarian crisis taxonomy, no disaster-type subdivision, no socio-political emergency category, and no compound-crisis structure. Empirical sampling confirms zero humanitarian content across 14 datapoints (US crime, UK culture, military affairs, celebrity features) [DATASET-D4, D5, D6, D8, D13]. The Ecuador deployment requires coverage of compound and socio-political crises scored by population impact and response-gap severity per the elicitation. This is a categorical mismatch on a HIGH-priority dimension.
CRISISLTLSUM evidence	CNN/DAILYMAIL evidence
[Q12] 'Moreover, we focus on the extraction of local crisis events. The term "local" indicates that an event is bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period.' (p.2)	[Q10] 'we seek to estimate the conditional probability $p(a c, q)$, where c is a context document, q a query relating to that document, and a the answer to that query.' (p.2)
[Q122] 'We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm.' (p.12)	[Q11] 'we wish to be able to exclude additional information, such as world knowledge gained from co-occurrence statistics, in order to test a model's core capability to detect and understand the linguistic relationships between entities in the context document.' (p.2)
[Q124] 'These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires...' (p.13)	[Q17] 'the focus of this paper is to provide a corpus for evaluating a model's ability to read and comprehend a single document, not world knowledge or co-occurrence.' (p.2)

[WEB-18] An Ecuador-specific Twitter monitoring system independently identified four distinct crisis categories — volcanic, telluric (seismic), fires, and climatological — diverging from the benchmark taxonomy	DATASET-D4 : British graphic artist feature — no crisis relevance
	DATASET-D6 : Frank Lloyd Wright/Salvador Dalí personality piece — entirely orthogonal to humanitarian taxonomy
	DATASET-D8 : US criminal home invasion case — no disaster content
	DATASET-D13 : US mass shooting — no humanitarian crisis category
	DATASET-D5 : Chinese toy recall — consumer safety, not humanitarian crisis

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

CRISISLTLSUM — 1/5 (Serious concern) [high confidence]	CNN/DAILYMAIL — 1/5 (Serious concern) [high confidence]
All 10,610 tweets in the benchmark were collected from seven US states and two Australian states during 2020–2021 [Q123] , selected explicitly because their event profiles match the benchmark's four crisis categories [Q124] . Tweet filtering relies on location candidates from US/Australian cities, towns, and 'famous neighborhoods' [Q30] and on English-curated keyword lists [Q126, Q127] . No Spanish-language, Latin American, indigenous-language, or humanitarian-organizational content is documented. The Ecuador deployment requires anchoring to parroquias, cantones, provincias, SNGR/SGR (renamed in 2024 [WEB-11]), Cruz Roja Ecuatoriana, IGEPN, INAMHI, and Kichwa/Shuar toponyms — none of which are present. The elicitation confirms US-default references would be 'operationally meaningless or misleading.' This is an instance-level content-validity violation that cannot be remediated through fine-tuning alone.	Content is exclusively drawn from CNN (~93k) and Daily Mail (~220k) English news articles 2007–2015 [Q4, Q9, Q14] , with all geographic and institutional references rooted in US/UK contexts [DATASET-D1, D2, D11, D16] . Empirical sampling shows references to Philadelphia, Miami, Alaska, London, Heathrow, Andersen AFB, the US CPSC, and UK monarchy — with zero Ecuadorian content. The deployment requires precise anchoring to Ecuador-specific entities at parroquia/cantón/provincia level (1,499 parroquias, 221+ cantones, 24 provincias [WEB-3, WEB-5, WEB-6]) and recognition of agencies like SGR, COE, Cuerpo de Bomberos, GAD, MSP, IGEPN [WEB-15, WEB-1] . Kichwa-origin place names appear in tweets but no NLP resources exist for Kichwa [WEB-8, WEB-9] . This is a categorical, HIGH-priority content mismatch.
CRISISLTLSUM evidence	CNN/DAILYMAIL evidence
[Q123] 'We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria.' (p.12)	[Q4] 'two new corpora of roughly a million news stories with associated queries from the CNN and Daily Mail websites.' (p.1)
[Q30] 'The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest.' (p.3)	[Q9] 'Articles were collected starting in April 2007 for CNN and June 2010 for the Daily Mail, both until the end of April 2015.' (p.2)
[Q126] 'We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past.' (p.13)	[Q14] 'We have collected 93k articles from the CNN and 220k articles from the Daily Mail websites.' (p.2)

[WEB-11] National disaster authority renamed from SGR to SNGR in 2023–2024	[WEB-3] Ecuador has 24 provincias requiring disambiguation
	[WEB-5] Ecuador has 221–222 cantones requiring sub-provincial anchoring
	[WEB-6] ~1,499 parroquias per INEC 2022 census
	[WEB-15] SGR is the primary Ecuadorian emergency management authority issuing COE resolutions
	[WEB-1] SGR open event data covers events 2010–2023 at province/cantón/parroquia level
	[WEB-8] Kichwa is extremely low-resource; first NLP dataset (Killkan) only released 2024
	DATASET-D1: Philadelphia/Miami police shooting — US institutional and geographic anchoring
	DATASET-D2: US Air Force F-15s and Russian bombers — US/Russian military, no Latin America
	DATASET-D11: Queen Elizabeth opens Heathrow Terminal 5 — UK monarchy and infrastructure
	DATASET-D16: US Consumer Product Safety Commission as institutional source — no SGR/COE analog

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

CRISISLTLSUM — 1/5 (Serious concern) [high confidence]	CNN/DAILYMAL — 1/5 (Serious concern) [high confidence]
The benchmark's preprocessing pipeline is English-specific throughout: AllenNLP NER trained on CoNLL03 [Q33], a BERT-based NER fine-tuned on English Twitter [Q35], OpenStreetMap location matching calibrated to US/Australian geography [Q36], English lemmatization and lowercasing [Q129], and English keyword matching [Q130]. The authors explicitly acknowledge: 'replicating the same process for other language or based on other locations may be hard because of such dependencies' [Q119]. The Ecuador deployment requires processing Ecuadorian Spanish with informal register, code-switching with Kichwa, non-standard orthography (the 2008 unified Kichwa alphabet is contested [WEB-1]), and Kichwa/Shuar toponyms for which no NLP resources exist [WEB-1, WEB-12]. No Kichwa pre-trained model or annotated corpus has been identified [domain_specific_notes]. BERT/BART tokenization is English-calibrated [Q86, Q141, Q143]. This is a fundamental external-validity violation: the entire signal-processing chain is built on assumptions that do not hold in the deployment.	Inputs are formal English news articles averaging 763 tokens with 27 entities per document [Q76], processed via tokenization into Deep LSTM encoders [Q40, Q42, Q43] with entity anonymization [Q18] that suppresses lexical surface form. Empirical sampling confirms uniformly formal journalistic English register with complete sentences and standard punctuation [DATASET-D14, D15, D18]. The deployment requires processing informal-to-semi-formal Ecuadorian Spanish tweets averaging ~280 characters, with non-standard orthography, missing diacritics, crisis abbreviations, and Kichwa entity tokens (per elicitation A4 and YAML). The HF metadata explicitly tags the dataset as `language:en` and `multilinguality:monolingual` (DATASET concern 1), confirming structural absence of Spanish content. The anonymization procedure [Q18] would explicitly suppress the dialect-specific entity signals critical for Ecuadorian anchoring. This is a categorical HIGH-priority form mismatch on language, register, length, and signal type.
CRISISLTLSUM evidence	CNN/DAILYMAL evidence

[Q33] 'we use a pre-trained neural model from AllenNLP (Gardner et al., 2018), trained on CoNLL03 (Tjong Kim Sang and De Meulder, 2003), to extract entities with types of people, location, and organization.' (p.3)	[Q18] 'we anonymise and randomise our corpora with the following procedure, a) use a coreference system to establish coreferents... b) replace all entities with abstract entity markers according to coreference; c) randomly permute these entity markers' (p.3)
[Q35] 'we further developed a BERT-based NER model tuned on Twitter data to detect location mentions.' (p.4)	[Q76] 'the average CNN validation document contained 763 tokens and 27 entities' (p.8)
[Q36] 'We use OpenStreetMap API to map location mentions to physical addresses.' (p.4)	DATASET-D14: Multi-clause formal English with institutional attribution (~30+ words per sentence)
[Q119] 'replicating the same process for other language or based on other locations may be hard because of such dependencies' (p.10)	DATASET-D15: Formal attributed quotation in journalistic prose
[Q129] 'we maintain both a lower-cased and a lemmitized version of the keywords and the tweet's text.' (p.13)	DATASET-D18: US English direct quotation, no register variation
[WEB-1] Ecuadorian indigenous languages have official alphabets but written use outside schools is minimal; 2008 unified Kichwa orthography contested	

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

CRISISLTLSUM — 2/5 (Significant gaps) [high confidence]	CNN/DAILYMAIL — 1/5 (Serious concern) [high confidence]
The output ontology has two parts: a binary timeline-membership label defined by three conjunctive criteria — relevant, informative, not repetitive [Q58] — and a four-axis summarization rubric (Coherence, Accuracy, Coverage, Overall quality, each 1–5) [Q106, Q151]. The criteria are abstract enough to be conceptually portable, but their operationalization is English- and US-context-specific: 'informative' is defined as facts vs. 'personal points of view' [Q58], an annotation norm reflecting US MTurk-worker intuitions; the Coherence rubric explicitly references 'English errors' and English-language readability [Q153]. The deployment's relevance criteria for humanitarian coordinators (road access status, informal community broadcasts, displacement indicators, IGEPN alert level changes, damnificado counts at parroquia level — see operational_context_notes) are structurally richer than the benchmark's binary fact-vs-opinion split. The elicitation rated OO as HIGH priority precisely because relevance for humanitarian coordinators differs structurally from US annotator norms. Score is 2 rather than 1 because the four-axis summary rubric is conceptually adaptable to Spanish summary evaluation, even though its English anchoring would need to be replaced.	The output ontology is a closed entity-classification schema: select the correct anonymized entity token from candidates appearing in the context [Q34], scored by accuracy [Q22, Q52]. No structured label taxonomy, no humanitarian category hierarchy, and no temporal/needs-evolution structure exists. The authors themselves flag that the model cannot scale to multi-sentence inference [Q66, Q67] or world knowledge [Q74], and that 'many queries requiring complex inference and long range reference resolution' remain unsolved [Q75]. The deployment requires temporally ordered narrative timelines with phase transitions, geographic anchoring at parroquia level, institutional attribution, and humanitarian needs taxonomy across access/health/WASH/shelter/food/protection/telecoms/SAR dimensions (YAML output_task_requirements). DATASET concern 4 confirms no temporal sequencing or needs-evolution structure in any output field. This is a categorical HIGH-priority structural mismatch.
CRISISLTLSUM evidence	CNN/DAILYMAIL evidence

[Q58] 'A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information.' (p.5)	[Q34] ' $p(a d, q) \propto \exp(W(a)g(d, q))$, s.t. $a \in V$, where V is the vocabulary' (p.4)
[Q106] 'Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality...' (p.8)	[Q66] 'the frame-semantic approach does not trivially scale to situations where several sentences, and thus frames, are required to answer a query.' (p.7)
[Q151] '...rate the quality of each of the eight summaries by four axes: coherence, accuracy, coverage, and overall quality.' (p.21)	[Q74] 'the incorporation of world knowledge and multi-document queries will also require the development of attention and embedding mechanisms whose complexity to query does not scale linearly' (p.8)
[Q153] 'A summary is coherent if when read by itself, it's easy to understand and free of English errors.' (p.21)	[Q75] 'There are still many queries requiring complex inference and long range reference resolution that our models are not yet able to answer.' (p.8)

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

CRISISLTLSUM — 1/5 (Serious concern) [high confidence]	CNN/DAILYMAIL — 1/5 (Serious concern) [high confidence]
Ground-truth labels were produced by three MTurk workers per timeline [Q56, Q70], explicitly restricted to 'countries where native English speakers are most likely to be found' [Q61]. The two reference summaries used for ROUGE scoring were chosen as those written by the two most-agreeing workers [Q87, Q88]. Summarization quality ratings came from a separate pool of five MTurk workers under the same location restriction [Q105, Q106]. No Spanish-speaking, Latin American, or humanitarian-practitioner annotators participated. The Ecuador deployment requires inter-operational validation by teams of Ecuadorian field coordinators, social analysts, and domain experts — explicitly not a crowd-sourced English-speaker pool (elicitation A3, OC priority HIGH). Inter-annotator agreement of 90.06% [Q77] and 91.77% against experts [Q79] is internally high but provides no evidence about Ecuadorian-stakeholder agreement. This is a direct convergent-validity violation: the labels reflect a different population's judgments than the deployment requires.	Ground-truth labels are generated entirely algorithmically by replacing entities in machine-extracted bullet summaries [Q3, Q16]; HF metadata explicitly tags `annotations_creators:no-annotation` (DATASET concern 5). No human annotation, no inter-annotator agreement, no demographic information, no humanitarian validation. The authors themselves acknowledge ambiguous queries that fail even under human evaluation [Q81, Q82, Q83, Q84], including geographic ambiguities like distinguishing administrative levels [Q82] — an acknowledged failure mode that maps directly to the Ecuador deployment's need for parroquia/cantón/provincia disambiguation. The deployment's authoritative quality standard is inter-operational team consensus across field coordinators, social analysts, and domain experts (elicitation A3, YAML). These two quality structures are categorically incompatible. HIGH-priority dimension.
CRISISLTLSUM evidence	CNN/DAILYMAIL evidence
[Q56] 'Each HIT contains three noisy timelines, and we collect annotations from 3 different workers on each.' (p.5)	[Q3] 'summary and paraphrase sentences, with their associated documents, can be readily converted to context–query–answer triples using simple entity detection and anonymisation algorithms.' (p.1)
[Q61] 'we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found.' (p.5)	[Q16] 'We construct a corpus of document–query–answer triples by turning these bullet points into Cloze [12] style questions by replacing one entity at a time with a placeholder.' (p.2)

[Q70] 'To make the ground-truth on whether each tweet is part of the timeline or not, we take the majority label among the three workers.' (p.6)	[Q81] 'ambiguous queries, where—at least following the anonymisation process—multiple entities are plausible answers even when evaluated manually.' (p.11)
[Q77] 'The average timeline-level agreement between those two workers is 90.06%.' (p.6)	[Q82] 'the query searches for an entity marker that describes a geographic location... Here it is unclear whether the placeholder refers to a part of town, town, region or country.' (p.11)
[Q105] 'we recruit a group of workers from MTurk with the same Location restriction (QC1)...' (p.8)	

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

CRISISLTLSUM — 2/5 (Significant gaps) [medium confidence]	CNN/DAILYMAIL — 1/5 (Serious concern) [high confidence]
Output modalities — binary classification labels and abstractive text summaries [Q82, Q84] — match the deployment's text-based requirements at the surface level (elicitation rated OF as MODERATE, the only non-HIGH dimension). However, evaluation is via ROUGE n-gram overlap against English reference summaries [Q98, Q99] and an English-language human rubric [Q153] , neither of which transfers cleanly to Spanish humanitarian reporting. Spanish-tokenization-aware ROUGE or multilingual BERTScore would be needed [evaluation_metric_adaptation], and UN OCHA / Red Cross situation reports follow specific genre conventions (numbered bullets, standardized headers, figures and percentages) that diverge from the abstractive summary style elicited in the benchmark [humanitarian_information_management]. Average timeline-level accuracy [Q91] as the extraction metric is generic and portable. The benchmark also does not penalize or evaluate language mismatch between input and output — so a system producing English summaries from Spanish tweets would not be flagged.	Output form is single-token classification accuracy on a closed entity vocabulary [Q34, Q52] , with hyperparameter tuning on validation [Q58, Q60, Q61] and attention heat-map interpretability [Q79, Q85, Q86, Q87, Q88] . The deployment requires temporally ordered, multi-sentence narrative timeline summaries with geographic-anchoring precision, institutional-attribution accuracy, temporal coherence, operational-relevance calibration, and bilingual-workflow support per YAML output_task_requirements. The HF dataset's summarization variant produces unordered abstractive bullet lists [DATASET-D1, D5] — also not a temporal timeline. Even at the closest structural overlap (abstractive summarization), the form is misaligned: no temporal sequencing, no phase transitions, no needs-typed multi-sentence narrative output. Memory constraints prevented architectural depth experiments [Q69] , and the authors flag the fixed-width hidden vector as an information bottleneck [Q45] . HIGH-priority dimension.
CRISISLTLSUM evidence	CNN/DAILYMAIL evidence
[Q82] 'we define two tasks: Timeline Extraction and Timeline Summarization.' (p.6)	[Q34] ' $p(a d, q) \propto \exp(W(a)g(d, q))$, s.t. $a \in V$, where V is the vocabulary' (p.4)
[Q91] 'For evaluation, we use the average timeline-level accuracy.' (p.7)	[Q52] 'we next evaluate these models on our reading comprehension corpora.' (p.6)
[Q98] 'We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004).' (p.8)	[Q45] 'The fixed width hidden vector forms a bottleneck for this information flow' (p.5)
[Q99] 'The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most...' (p.8)	[Q69] 'Memory constraints prevented us from experimenting with deeper Attentive Readers.' (p.7)
[Q153] 'A summary is coherent if when read by itself, it's easy to understand and free of English errors.' (p.21)	DATASET-D1 : Four-bullet US crime summary — no temporal evolution